

Assignment 3

Regression: Recurrent Neural Network (LSTM)

SCANIA Component X Dataset - Predictive Maintenance

Course: AI & Neural Networks

University: Astana IT University (AITU)

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1. Introduction

This report presents Stage 3 of the SCANIA Component X project, where a Long Short-Term Memory (LSTM) neural network was used for regression on sequential vehicle data. The goal is to improve on the dense MLP/DNN models from Assignment 2 by explicitly modeling temporal history.

The notebook for this assignment was implemented in English and executed successfully. The final notebook, trained LSTM model, and exported CSV outputs were used as the basis for this PDF report.

2. Data Preparation and Sequence Construction

2.1 Source Data

The processed dataset from Assignment 1 was reused. It contains 50,000 rows and 123 columns. After excluding vehicle_id, time_step, class_label, length_of_study_time_step, and time_of_day, 118 numerical input features remained.

2.2 Sequence Constraint in the Sampled Dataset

The sampled Assignment 1 dataset contains 18,852 vehicles, but most per-vehicle sequences are very short. The maximum observed sequence length is only 16 rows, which is shorter than the requested experimental windows of 24 and 48. To keep vehicle boundaries intact and still test larger windows, right-padded history windows and a Masking layer were used.

2.3 Split and Standardization

The data was sorted by vehicle_id and time_step and split chronologically using the same 70/15/15 strategy as Assignment 2. StandardScaler was fit on the training partition only, then applied to validation and test data to prevent leakage.

- Train rows: 35,000
- Validation rows: 7,500
- Test rows: 7,500
- Sequence counts for each tested window: 22,473 train / 4,511 validation / 4,164 test

3. LSTM Model

3.1 Architecture

The recurrent regression model used the following architecture:

Input(window, 118) -> Masking(0.0) -> LSTM(64) -> Dense(32, ReLU) -> Dense(1, Linear)

3.2 Training Configuration

- Optimizer: Adam
- Loss function: Mean Squared Error (MSE)
- Metric: Mean Absolute Error (MAE)
- Batch size: 128
- Max epochs: 15
- EarlyStopping: patience=3, restore_best_weights=True
- Tested history windows: 12, 24, 48

4. Experimental Results

The best production model was selected by validation RMSE, not by test RMSE. This is the correct selection rule for model development. In this run, the saved best model corresponded to a window size of 12, while the lowest observed test RMSE happened at window 48.

Window	Epochs	Val RMSE	Val MAE	Test RMSE	Test MAE
12	11	51.557	39.089	59.706	49.923
24	11	52.354	39.590	60.643	50.700
48	12	51.889	40.103	54.037	44.037

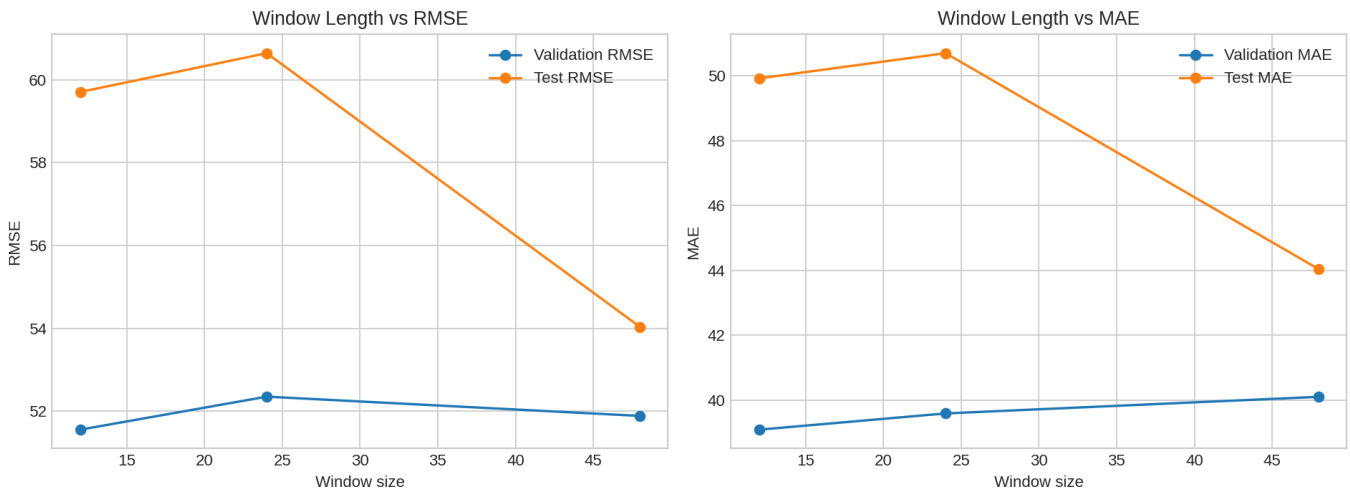


Figure 1: Validation and test metrics across history window sizes

5. Prediction Quality

The saved best LSTM model achieved a test RMSE of 59.706 and a test MAE of 49.923. Prediction quality is illustrated both as a line plot for the first 250 test samples and as a scatter plot against the identity line.

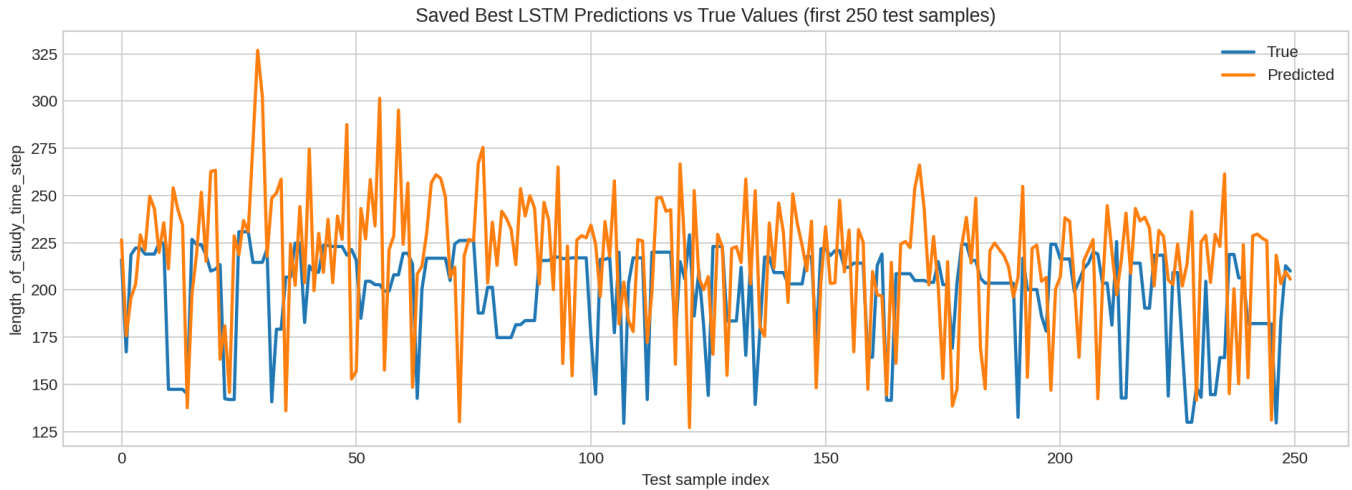


Figure 2: Saved best LSTM predictions vs true values

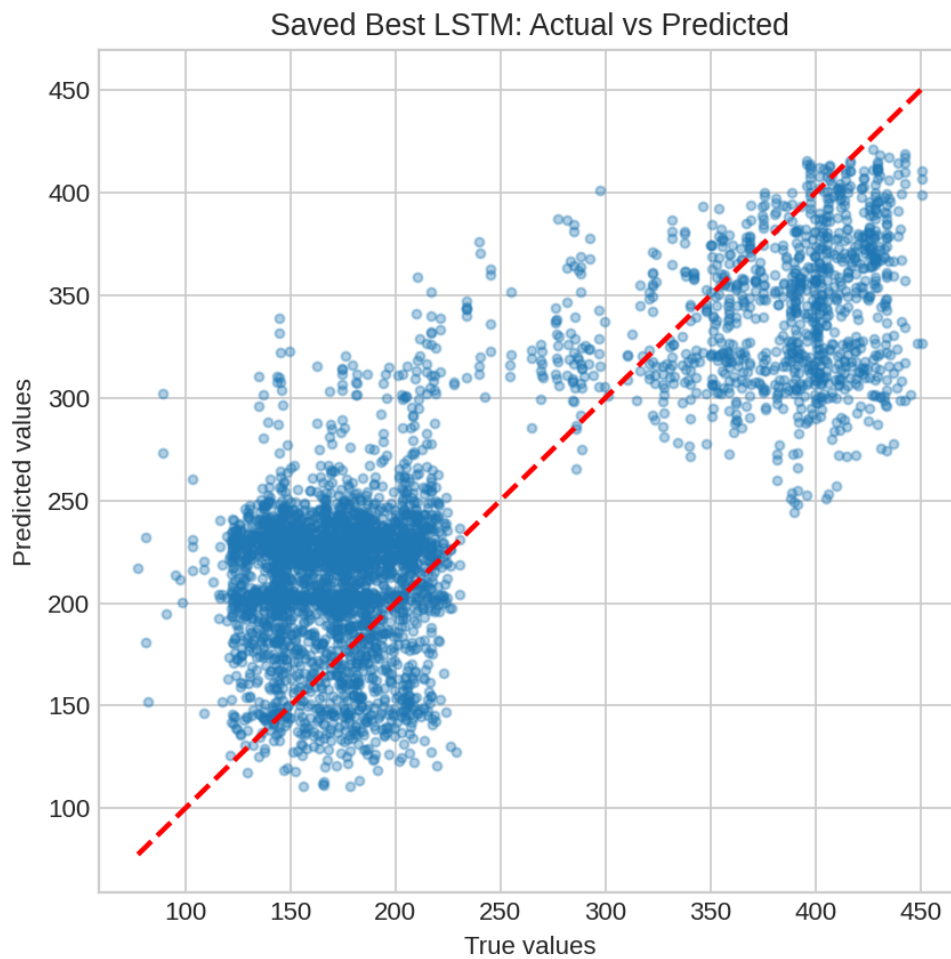


Figure 3: Actual vs predicted values for the saved best LSTM

6. Comparison with Assignment 2

Assignment 2 reported the best dense model as a DNN with test RMSE 67.6085 and test MAE 55.1771. Compared with that baseline, the saved LSTM model selected by validation RMSE improved both metrics.

- Saved best LSTM (window=12) test RMSE: 59.706
- Saved best LSTM (window=12) test MAE: 49.923
- Improvement over Stage 2 DNN in RMSE: 7.902
- Improvement over Stage 2 DNN in MAE: 5.254
- Lowest observed Stage 3 test RMSE overall: 54.037 at window=48

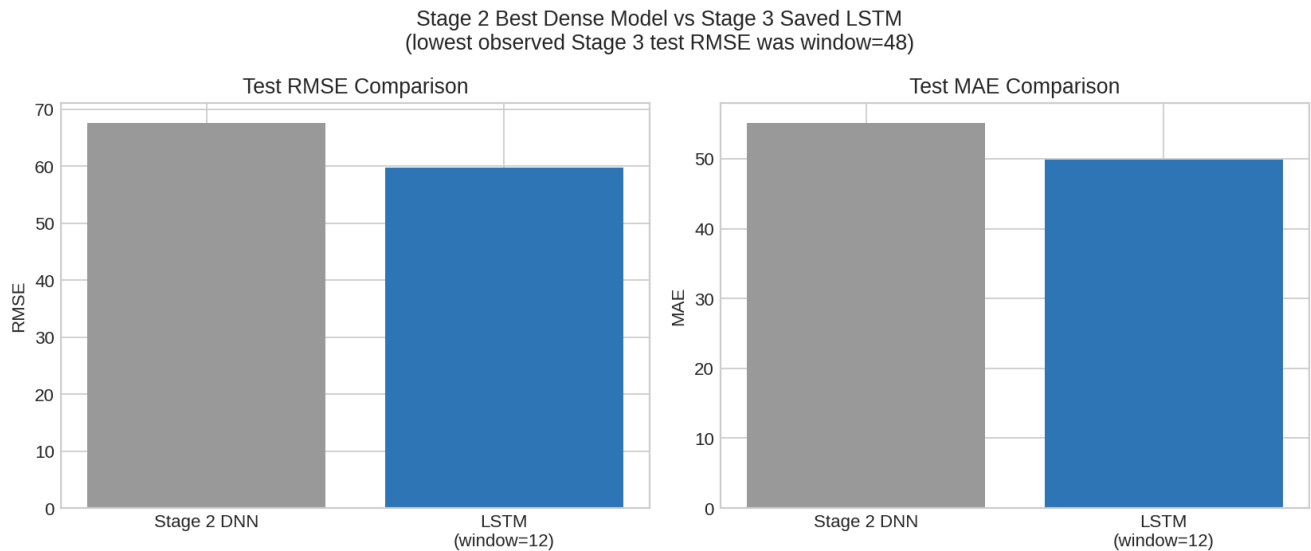


Figure 4: Stage 2 dense baseline vs Stage 3 saved LSTM

7. Conclusions

1. LSTM models successfully incorporated temporal context and improved on the best dense model from Assignment 2.
2. Because the sampled dataset contains short per-vehicle histories, padding plus masking was required to test larger windows without breaking chronological integrity.
3. The saved production model was selected by validation RMSE (window 12), while the lowest observed test RMSE in the window experiment occurred at window 48. This distinction was documented explicitly in the notebook and in this report.
4. The main deliverables for Assignment 3 are the executed Jupyter notebook, the saved model lstm_best.h5, the window-level results CSV, and this PDF report.